

# The Korteweg–de Vries (KdV) Equation

# The KdV Equation

## Standard form:

$$u_t + 6uu_x + u_{xxx} = 0$$

## Interpretation:

- ▶  $u(x, t)$ : wave profile
- ▶  $6uu_x$ : nonlinear steepening
- ▶  $u_{xxx}$ : dispersive smoothing

## Context:

- ▶ Shallow water waves
- ▶ Balance of nonlinearity and dispersion

# Conservation Laws

The KdV equation has an infinite number of conserved quantities.

**First three:**

$$M = \int_{\mathbb{R}} u \, dx$$

$$P = \int_{\mathbb{R}} u^2 \, dx$$

$$E = \int_{\mathbb{R}} \left( \frac{1}{2} u_x^2 - u^3 \right) dx$$

We focus on the energy functional:

$$E[u] = \int \left( \frac{1}{2} u_x^2 - u^3 \right) dx$$

# Energy Conservation

Compute the time derivative:

$$\frac{dE}{dt} = \int (u_x u_{xt} - 3u^2 u_t) dx$$

Use the PDE:

$$u_t = -6uu_x - u_{xxx}$$

Substitute and integrate by parts (assuming decay at infinity).

**Result:**

$$\frac{dE}{dt} = 0$$

**Conclusion:** Energy is conserved.

# Sketch of Energy Derivation

Key steps:

- ▶ Replace  $u_t$  using KdV
- ▶ Terms become total derivatives:

$$\int u_x u_{xxxx} dx, \quad \int u^2 u_{xxx} dx$$

- ▶ Integrate by parts repeatedly
- ▶ Boundary terms vanish (rapid decay or periodic BCs)

**Takeaway:**

$$\frac{d}{dt} \int \left( \frac{1}{2} u_x^2 - u^3 \right) dx = 0$$

# Existence and Uniqueness

## Theorem (Local Well-Posedness):

Let  $s > \frac{3}{2}$  and  $u_0 \in H^s(\mathbb{R})$ . Then there exists  $T > 0$  and a unique solution

$$u \in C([0, T]; H^s(\mathbb{R}))$$

to the KdV equation.

## Sketch of proof:

- ▶ Rewrite as evolution equation:

$$u_t = -6uu_x - u_{xxx}$$

- ▶ Linear part generates a unitary group
- ▶ Use fixed point argument (Picard iteration)
- ▶ Sobolev embedding controls  $uu_x$

# Global Existence

## **Theorem (Global existence):**

For sufficiently smooth initial data, solutions exist for all  $t > 0$ .

## **Idea:**

- ▶ Use conservation of  $\|u\|_{L^2}$
- ▶ Use energy conservation
- ▶ Prevent blow-up via a priori bounds

KdV is completely integrable, giving strong global control.

# Travelling Wave Ansatz

Seek solutions of the form:

$$u(x, t) = f(x - ct)$$

Let  $\xi = x - ct$ .

Then:

$$u_t = -cf', \quad u_x = f', \quad u_{xxx} = f'''$$

Substitute into KdV:

$$-cf' + 6ff' + f''' = 0$$

# Reduction to ODE

Integrate once:

$$-cf + 3f^2 + f'' = A$$

For localized solutions, take  $A = 0$ :

$$f'' = cf - 3f^2$$

Multiply by  $f'$  and integrate:

$$\frac{1}{2}(f')^2 = \frac{c}{2}f^2 - f^3$$

# Soliton Solution

Solve:

$$(f')^2 = cf^2 - 2f^3$$

This yields:

$$f(\xi) = \frac{c}{2} \operatorname{sech}^2\left(\frac{\sqrt{c}}{2}\xi\right)$$

**Final travelling wave:**

$$u(x, t) = \frac{c}{2} \operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct)\right)$$

**Properties:**

- ▶ Localized
- ▶ Stable
- ▶ Shape preserved under interaction

# Summary

- ▶ KdV equation:

$$u_t + 6uu_x + u_{xxx} = 0$$

- ▶ Conserved energy:

$$E = \int \left( \frac{1}{2} u_x^2 - u^3 \right) dx$$

- ▶ Well-posed in Sobolev spaces
- ▶ Admits soliton solutions

## Second Conserved Quantity

We derive conservation of

$$P[u] = \int_{\mathbb{R}} u^2 dx$$

Compute:

$$\frac{d}{dt} \int u^2 dx = 2 \int uu_t dx$$

Use the KdV equation:

$$u_t = -6uu_x - u_{xxx}$$

Thus:

$$\frac{d}{dt} \int u^2 dx = -12 \int u^2 u_x dx - 2 \int uu_{xxx} dx$$

# Derivation (Integration by Parts)

We handle each term separately.

**First term:**

$$\int u^2 u_x dx = \frac{1}{3} \int (u^3)_x dx = 0$$

**Second term:**

$$\int u u_{xxx} dx = - \int u_x u_{xx} dx = -\frac{1}{2} \int (u_x^2)_x dx = 0$$

(All boundary terms vanish for decaying or periodic solutions.)

**Conclusion:**

$$\frac{d}{dt} \int u^2 dx = 0$$

**So:**  $P[u] = \int u^2 dx$  is conserved.

# Hierarchy of Conserved Quantities

The KdV equation has an infinite sequence of conserved quantities.

**First three:**

$$M[u] = \int_{\mathbb{R}} u \, dx \quad (\text{mass})$$

$$P[u] = \int_{\mathbb{R}} u^2 \, dx \quad (\text{momentum})$$

$$E[u] = \int_{\mathbb{R}} \left( \frac{1}{2} u_x^2 - u^3 \right) dx \quad (\text{energy})$$

**Structure:**

- ▶ Each conserved quantity controls higher regularity
- ▶ They arise from an underlying structure structure

**Key takeaway:** KdV has infinitely many conservation laws

This is a hallmark of *complete integrability*.

# Lax Pair Formulation

A key feature of KdV is that it admits a **Lax pair**.

Define operators:

$$L = -\partial_x^2 + u(x, t), \quad P = -4\partial_x^3 + 3(u\partial_x + \partial_x u)$$

Then the KdV equation is equivalent to:

$$\frac{dL}{dt} = [P, L] = PL - LP$$

## Interpretation:

- ▶ Evolution is isospectral
- ▶ Eigenvalues of  $L$  are time-invariant

# Verification of the Lax Pair

Apply both sides to a test function  $\psi$ .

Left-hand side:

$$\frac{d}{dt}(L\psi) = u_t\psi + L\psi_t$$

Right-hand side:

$$[P, L]\psi = P(L\psi) - L(P\psi)$$

A direct computation shows:

$$[P, L] = (6uu_x + u_{xxx}) \cdot (\text{multiplication operator})$$

Thus:

$$\frac{dL}{dt} = [P, L] \iff u_t + 6uu_x + u_{xxx} = 0$$

# Spectral Problem

Consider the Schrödinger operator:

$$L\psi = \lambda\psi \quad \Longleftrightarrow \quad -\psi'' + u(x, t)\psi = \lambda\psi$$

**Key idea:**

- ▶  $u(x, t)$  acts as a potential
- ▶ The spectrum of  $L$  encodes the solution

Under KdV flow:

$\lambda$  is constant in time

**This is the basis of inverse scattering.**

# Direct Scattering

Given initial data  $u(x, 0)$ :

Solve:

$$-\psi'' + u(x, 0)\psi = \lambda\psi$$

Extract **scattering data**:

- ▶ Reflection coefficient  $R(\lambda)$
- ▶ Discrete eigenvalues  $\{\lambda_n\}$
- ▶ Norming constants

**Interpretation:**

- ▶ Continuous spectrum  $\rightarrow$  radiation
- ▶ Discrete spectrum  $\rightarrow$  solitons

# Time Evolution of Scattering Data

Under KdV evolution:

- ▶ Eigenvalues  $\lambda_n$  are constant
- ▶ Reflection coefficient evolves simply:

$$R(\lambda, t) = R(\lambda, 0) e^{8i\lambda^{3/2}t}$$

**Key simplification:**

- ▶ Complicated PDE  $\rightarrow$  simple phase evolution

# Inverse Scattering

Reconstruct  $u(x, t)$  from scattering data.

Steps:

1. Compute scattering data from  $u(x, 0)$
2. Evolve data explicitly in time
3. Solve inverse problem (Gel'fand–Levitan–Marchenko equation)

**Outcome:**

- ▶ Exact solution for all  $t$
- ▶ Solitons emerge from discrete spectrum

# Solitons and the Spectrum

Each negative eigenvalue corresponds to a soliton.

For a single eigenvalue:

$$\lambda = -\kappa^2$$

This produces:

$$u(x, t) = \frac{c}{2} \operatorname{sech}^2 \left( \frac{\sqrt{c}}{2} (x - ct) \right)$$

## Interpretation:

- ▶ Discrete spectrum  $\leftrightarrow$  solitons
- ▶ Multi-soliton solutions from multiple eigenvalues

## Summary: Integrability

- ▶ Lax pair  $\Rightarrow$  isospectral flow
- ▶ Infinite conserved quantities
- ▶ Scattering data evolves linearly
- ▶ Nonlinear PDE solved via linear spectral theory

### Conclusion:

KdV is a completely integrable system

# Nonlinear Schrödinger Equation (NLS)

**Standard (focusing) form:**

$$i\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0$$

**Interpretation:**

- ▶  $\psi(x, t)$ : complex wave envelope
- ▶  $\psi_{xx}$ : dispersion
- ▶  $2|\psi|^2\psi$ : nonlinear self-phase modulation

**Connection to KdV:**

- ▶ Both are completely integrable
- ▶ Both admit soliton solutions
- ▶ Both arise from balance of nonlinearity and dispersion

# NLS in Optical Communication

## Physical setting:

- ▶ Light propagation in optical fibers
- ▶  $\psi(x, t)$  represents a slowly varying electric field envelope

## Key effects:

- ▶ Dispersion spreads pulses
- ▶ Nonlinearity focuses them

## Solitons:

- ▶ Stable pulse solutions
- ▶ Maintain shape over long distances
- ▶ Resist dispersion-induced broadening

## Application:

- ▶ Soliton-based communication systems
- ▶ Long-distance transmission with minimal distortion

# KdV on a Periodic Domain

We consider:

$$u_t + 6uu_x + u_{xxx} = 0, \quad x \in [0, L], \quad \text{periodic}$$

**Fourier representation:**

$$u(x, t) = \sum_{k \in \mathbb{Z}} \hat{u}_k(t) e^{ik\kappa x}, \quad \kappa = \frac{2\pi}{L}$$

**Key observations:**

- Derivatives become multipliers:

$$\widehat{u_x} = ik\kappa \hat{u}_k, \quad \widehat{u_{xxx}} = (ik\kappa)^3 \hat{u}_k$$

- Linear part is diagonal in Fourier space
- Nonlinearity is local in physical space

# Operator Splitting

Write KdV as:

$$u_t = \underbrace{-u_{xxx}}_{\text{linear } L(u)} + \underbrace{-6uu_x}_{\text{nonlinear } N(u)}$$

We approximate evolution over  $\Delta t$  by splitting:

$$e^{\Delta t(L+N)} \approx e^{\frac{\Delta t}{2}L} e^{\Delta t N} e^{\frac{\Delta t}{2}L}$$

## Strang splitting:

- ▶ Second-order accurate in time
- ▶ Linear part handled exactly in Fourier space
- ▶ Nonlinear part handled in physical space

# Linear Step (Fourier Space)

Solve:

$$u_t = -u_{xxx}$$

In Fourier space:

$$\partial_t \hat{u}_k = -(ik\kappa)^3 \hat{u}_k$$

**Exact solution:**

$$\hat{u}_k(t + \Delta t) = e^{-(ik\kappa)^3 \Delta t} \hat{u}_k(t)$$

**Implementation:**

- ▶ FFT  $\rightarrow$  multiply by phase factor  $\rightarrow$  inverse FFT

# Nonlinear Step (Physical Space)

Solve:

$$u_t = -6uu_x$$

## Method:

- ▶ Compute  $u_x$  spectrally (FFT differentiation)
- ▶ Form  $-6uu_x$  in physical space
- ▶ Advance with explicit step (e.g., forward Euler or RK)

## Remark:

- ▶ Often a simple explicit step suffices due to small  $\Delta t$
- ▶ Can use RK2/RK4 for improved stability

# Aliasing and Dealiasing

Nonlinear term  $uu_x$  introduces higher frequencies.

## Issue:

- ▶ Mode folding (aliasing) corrupts solution

## Fix (2/3 rule):

- ▶ Zero out top 1/3 of Fourier modes
- ▶ Apply after each nonlinear evaluation

## Result:

- ▶ Improved stability and accuracy

# Split-Step Fourier Algorithm

At each time step:

1. Half linear step (Fourier space)
2. Full nonlinear step (physical space)
3. Half linear step (Fourier space)

## Advantages:

- ▶ Spectral accuracy in space
- ▶ Efficient via FFT ( $O(N \log N)$ )
- ▶ Excellent for periodic problems

# Pseudocode (Split-Step Fourier KdV)

Given:  $u(x, 0)$  on grid  $x_j$ , timestep  $\Delta t$ , modes  $k$

Precompute:

$$\lambda_k = -(ik\kappa)^3 \quad \text{and} \quad E_k = e^{\lambda_k \Delta t/2}$$

For  $n = 0, 1, 2, \dots$ :

1.  $\hat{u} \leftarrow \text{FFT}(u)$
2.  $\hat{u} \leftarrow E_k \cdot \hat{u}$  (half linear step)
3.  $u \leftarrow \text{IFFT}(\hat{u})$
4. Compute  $u_x$  via FFT differentiation
5.  $u \leftarrow u - \Delta t \cdot 6uu_x$  (nonlinear step)
6. Dealias high-frequency modes (2/3 rule)
7.  $\hat{u} \leftarrow \text{FFT}(u)$
8.  $\hat{u} \leftarrow E_k \cdot \hat{u}$  (half linear step)
9.  $u \leftarrow \text{IFFT}(\hat{u})$